

The Thermodynamic Identity Governing the Virial Theorem: Physical Identification of K and Evidence Across 37 Orders of Magnitude

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The virial theorem, established by Clausius in 1870, states that for any stable bound system under a $1/r$ potential, $2\langle K \rangle + \langle V \rangle = 0$. The theorem has been verified across 37 orders of magnitude, but the physical meaning of K beyond “average kinetic energy of bound particles” has remained unexplained. We show that the virial theorem and Landauer’s principle are two expressions of the same thermodynamic identity. The first law gives the heat released at the virialization boundary, $Q = -E_f$; the second law and Landauer’s principle give $E_{\text{Landauer}} = Q$; the equilibrium condition of the $1/r$ potential gives $-E_f = \langle K \rangle$. Together these identify $\langle K \rangle$ as the Landauer cost of the virialization transition, encoded onto the nearest available surface as entropy at cost $k_B T \ln 2$ per bit. The virial theorem is the mechanical expression of this thermodynamic identity. The identity is consistent with data across 37 orders of magnitude in physical scale. We propose this thermodynamic identity constitutes a fundamental law of physics of which the virial theorem is the mechanical special case.

INTRODUCTION

The virial theorem has governed bound systems since Clausius [1]:

$$2\langle K \rangle + \langle V \rangle = 0, \quad \langle K \rangle = \frac{1}{2}|\langle V \rangle|, \quad (1)$$

verified from the hydrogen atom (10^{-10} m) to galaxy clusters (10^{23} m)—37 orders of magnitude. It is one of the most broadly tested results in physics.

Yet a question has gone unanswered for 154 years: what is K ?

Clausius derived Eq. (1) from Newton’s second law. The derivation is mechanical and describes the equilibrium state. It is silent on the thermodynamic content of the transition that produces the equilibrium, and silent on the physical identity of the energy released during that transition. The thermodynamics of gravitational systems has been studied extensively [13, 14], but what has not been shown is that the virial theorem and Landauer’s principle [2] are two expressions of the same identity, and that this identity provides the physical meaning of $\langle K \rangle$ as the Landauer cost of the virialization transition itself.

We show this here. The argument proceeds in four steps from principles every physicist accepts: the first law, the second law, Landauer’s principle, and the equilibrium condition of the $1/r$ potential.

DERIVATION

Setup.—Consider a system of total mass M transitioning from an unbound state to a stable bound state under a $1/r$ potential. The natural reference state is the unbound system at rest at infinity, with $E_i = 0$. The bound equilibrium state has total energy $E_f = \langle K \rangle + \langle V \rangle < 0$.

Step 1 (First Law).—The transition from unbound to bound is irreversible: a virialized system does not

spontaneously return to the unbound state. The heat released irreversibly to the environment is

$$Q = E_i - E_f = -(\langle K \rangle + \langle V \rangle). \quad (2)$$

This is exact. $Q > 0$ because $E_f < 0$ for any stable bound state.

Step 2 (Second Law).—The irreversible transition produces entropy:

$$\Delta S = Q/T, \quad (3)$$

where T is the temperature of the bound system.

Step 3 (Landauer).—By Landauer’s principle [2], the minimum thermodynamic cost of producing entropy ΔS irreversibly is $E_{\text{Landauer}} = T \Delta S$. Substituting Eq. (3):

$$E_{\text{Landauer}} = T \times \frac{Q}{T} = Q. \quad (4)$$

The temperature cancels exactly. The Landauer cost equals the heat released at the boundary, independently of the scale or temperature of the system.

Step 4 (1/r equilibrium condition).—Steps 1–3 establish $E_{\text{Landauer}} = Q = -E_f$. For a potential $V \propto r^{-1}$, Euler’s theorem for homogeneous functions of degree -1 requires $2\langle K \rangle + \langle V \rangle = 0$ at stable bound equilibrium, which gives $-E_f = \langle K \rangle$. This is the equilibrium condition of the $1/r$ potential—the condition that defines what it means for such a system to be stably bound. Combining with Eqs. (2) and (4):

$$\boxed{\langle K \rangle = Q = T \Delta S = E_{\text{Landauer}} = \frac{1}{2}|\langle V \rangle|}. \quad (5)$$

Remark.—This is not a derivation of the virial theorem from thermodynamics alone. It is the demonstration that the virial theorem and Landauer’s principle are two expressions of the same identity: the $1/r$ equilibrium condition gives the mechanical ratio; Landauer’s principle

identifies the physical content of that ratio as the irreversible thermodynamic cost of the virialization transition. The $1/2$ partition is the unique ratio at which both are simultaneously satisfied.

THE THERMODYNAMIC IDENTITY: FORMAL STATEMENT

As a system crosses from an unbound to a bound state under a $1/r$ potential, it releases heat Q irreversibly at the boundary. The first law, the second law, Landauer’s principle, and the equilibrium condition of the $1/r$ potential together require $\langle K \rangle = Q = T \Delta S = E_{\text{Landauer}}$. The heat Q is encoded onto the nearest available surface as entropy at thermodynamic cost $k_B T \ln 2$ per bit. The virial theorem $2\langle K \rangle + \langle V \rangle = 0$ is the mechanical expression of this condition. A stable bound state exists if and only if both are satisfied simultaneously.

The $1/2$ partition is not a free choice. It is the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied for a $1/r$ potential. Scale-invariance follows directly from the scale-invariance of Landauer’s principle: the principle applies wherever an irreversible process produces entropy, at any scale or temperature.

EVIDENCE ACROSS 37 ORDERS OF MAGNITUDE

Table I presents the evidence for the thermodynamic identity across every physical domain in which bound systems under $1/r$ potentials have been tested. The range spans 37 orders of magnitude in physical scale.

TABLE I. Evidence for the thermodynamic identity $\langle K \rangle = Q = \frac{1}{2}|\langle V \rangle|$ across 37 orders of magnitude. [†]17 independent MCMC chains, zero free parameters.

System	Scale	$K/ V $	Result
Electron m_e	λ_C	f.p.	6.6 ppm [20]
Atoms (20)	10^{-10} m	$T/ V $	1.0000 [6, 7]
Molecules (10)	10^{-9} m	$T/ V $	1.0000 [6, 7]
Equipartition	thermal	$k_B T/2$	exact
Solar	10^9 m	$K/ U $	$\approx 0.5, 10\%$
Chandrasekhar	10^7 m	M_{Ch}	$1.44 M_\odot$ [3]
Clusters	10^{23} m	$K/ U $	$\approx 0.5, 20\%$
N-body (6)	10^{22-25} m	$2K/ U $	0.815 ± 0.025 [9–12, 15, 16]
$\beta_m^\dagger$	10^{26} m	β_m/Ω_m	0.1583 ± 0.0033 [5, 18]

The electron mass entry uses the thermodynamic identity applied at the Compton scale: rest energy equals the Landauer cost of maintaining the particle’s decoherence loop at the Gibbons-Hawking temperature of the cosmic horizon [20]. The prediction uses only $\hbar$, H_0 , m_P , and α with no fitted parameters.

The atomic and molecular entries use published Hartree-Fock limit energies [6, 7]: the mean virial ratio $\eta = T/|V| = 1.0000000000$ with standard deviation zero across all 20 elements and 10 molecules. For converged Hartree-Fock wavefunctions this is analytically exact, not a numerical coincidence.

The N-body entry compiles six independent simulation studies [9–12, 15, 16] spanning 25 years of published literature, covering halo masses 10^{10} – $10^{15} M_\odot$ and redshifts $z = 0$ – 1 , using different simulation codes and analysis pipelines. The predicted $\langle \eta_{\text{vir}} \rangle = 0.81$ follows from the virial partition and the published collapsed fraction $f_{\text{coll}} = 0.62 \pm 0.03$ [8]: $\eta_{\text{vir}} = 1/(2f_{\text{coll}}) = 0.81$. The combined measurement $\Omega_m \times f_{\text{coll}} \times \langle \eta_{\text{vir}} \rangle = 0.315 \times 0.62 \times 0.815 = 0.159 \pm 0.010$ agrees with the thermodynamic identity prediction $\beta_m = \Omega_m/2 = 0.1575$ to 1.0% with no free parameters.

The cosmological entry instantiates the thermodynamic identity at cosmic scales via the Informational Actualization Model (IAM) [17], deriving $\beta_m = \Omega_m/2 = 0.1575$ from the same $1/2$ partition with no fitted parameters. Seventeen independent MCMC chains against the full Planck 2018 likelihood [5] return $\beta_m = 0.1583 \pm 0.0033$, consistent at 0.2σ . The MCMC sampler has no knowledge of the thermodynamic identity; the posterior arrives at this value because the data are consistent with it.

RELATIONSHIP TO JACOBSON 1995

Jacobson [4] derived the Einstein field equations from $\delta Q = T \delta S$ applied to the Rindler horizon of an accelerated observer. That derivation showed that the field equations are thermodynamic in origin. The present work applies the same identity $Q = T \Delta S$ to the virialization boundary and establishes that the virial theorem is its mechanical expression. Both start from the same thermodynamic identity. The present work extends the Jacobson program from the field equations governing curved spacetime to the equilibrium condition governing bound states within that spacetime.

CONCLUSION

The virial theorem $2\langle K \rangle + \langle V \rangle = 0$ has been verified across 37 orders of magnitude for 154 years without a physical identification of $\langle K \rangle$ beyond “average kinetic energy of bound particles.”

The first law, the second law, Landauer’s principle, and the equilibrium condition of the $1/r$ potential together establish $\langle K \rangle = Q = T \Delta S = E_{\text{Landauer}}$: the kinetic energy of a virialized system is the Landauer cost of the virialization transition, encoded permanently as entropy on the nearest available surface. The virial theorem is the mechanical expression of this thermodynamic identity.

The $1/2$ partition is not a numerical coincidence. It is the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied for a $1/r$ potential. Its appearance across 37 orders of magnitude is a consequence of the scale-invariance of Landauer's principle and the universality of the $1/r$ potential form.

We propose that this thermodynamic identity constitutes a fundamental law of physics of which the virial theorem is the mechanical special case.

The thermodynamic framework of T. Jacobson [4] is the foundation upon which this work is built. All MCMC chains and supporting papers are publicly archived at <https://github.com/hmahaffeyges/IAM-Validation> (Zenodo DOI: 10.5281/zenodo.18702042).

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